Recurrence Diagnostics for Econometric Stress Episodes: A Methods Paper with a Cluster-Adjusted Stress Functional

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Abstract

Standard stress diagnostics in applied econometrics typically measure the magnitude of variation, tail losses, or event frequency, but not the temporal organization of extreme episodes. That omission matters whenever policy or risk interpretation depends on whether extremes arrive as isolated shocks or as clustered stress. This paper develops a reproducible recurrence-oriented framework for stress diagnostics and introduces a cluster-adjusted stress functional that combines exceedance probability with tail clustering. The paper has two goals. First, it provides a serious methods contribution: a formal probabilistic setup, explicit estimators for exceedance incidence, return-time behavior, runs-based clustering, and multivariate stress activation, and a reproducible implementation in the accompanying repository. Second, it adds a narrow theoretical contribution: basic ordering and plug-in consistency results for the proposed cluster-adjusted functional under standard high-threshold regularity conditions. Synthetic evidence generated by the repository shows that recurrence diagnostics separate benchmark processes that would be inadequately summarized by volatility alone. In the current artifact set, the estimated extremal indices for two stress benchmarks are 0.673 and 0.780, the return-time distribution has median 14 and 90th percentile 40.2, and rolling volatility is elevated during exceedance periods without exhausting the clustering information. The paper makes modest claims: it does not offer new asymptotic theory for extremal-index estimation itself, nor does it claim structural macroeconomic identification. Its contribution is a mathematically explicit and computationally reproducible bridge between extreme-value diagnostics and econometric stress analysis.

Keywords: extreme values, extremal index, recurrence, return times, stress testing, time series econometrics

1 Introduction

The motivating problem is straightforward. In many macro-financial and econometric applications, a "stress episode" is treated as a period in which a variable becomes large in magnitude, volatile, or both. But a stress episode is not defined only by how large observations become. It is also defined by how they recur. A series that produces isolated tail observations conveys a different kind of fragility from a series that produces clustered extremes, even if the two series have comparable event counts or similar rolling volatility.

That distinction is conceptually obvious and methodologically underused. Standard variance-based diagnostics, from ARCH and GARCH models onward, are designed to summarize time-varying second moments [8, 4]. Tail-risk and systemic-risk measures such as VaR, CoVaR, expected shortfall, and capital-shortfall metrics focus on the severity of bad states and their cross-sectional comovement [1, 2, 5, 3].

Yet the temporal spacing of threshold exceedances is a different object. Extreme-value theory for dependent sequences treats exceedance clustering and return times as first-order features of the data-generating process rather than as narrative afterthoughts [12, 7, 6]. The extremal index provides a canonical summary of tail clustering [14, 9, 15]. Related recurrence ideas also appear in the study of hitting times and rare events for stochastic and dynamical systems [11, 10, 13]. What is comparatively rare in applied econometrics is a workflow that makes these objects operational in a transparent, reproducible way and places them next to familiar volatility baselines rather than outside the analysis.

The gap addressed here is therefore narrower, and in some ways more demanding, than a generic shortage of methods. Extreme-value theory already provides a mature language for thinking about exceedance clustering, return times, and dependence in the tail, while econometric practice already offers highly developed tools for modeling volatility, downside risk, and financial stress. What remains comparatively underdeveloped is a framework in which recurrence itself is treated as an object of stress measurement and is analyzed alongside, rather than outside, standard econometric summaries. That is the problem taken up in this paper. I formalize a recurrence-based stress framework built from exceedance incidence, return-time behavior, runs-based clustering, multivariate stress activation, and volatility baselines; I implement those objects in a transparent computational pipeline that records intermediate artifacts rather than only final graphics; and I introduce a cluster-adjusted stress functional that distinguishes between stress processes with similar incidence but different temporal concentration of extremes. The theoretical contribution is intentionally narrow, but it is not decorative: the functional is given explicit probabilistic content, basic ordering properties, and plug-in consistency under standard regularity conditions. The broader aim is not to compete with the deep asymptotic theory of dependent extremes, but to make part of that theory usable in the setting of applied stress diagnosis.

The paper's empirical claims remain modest. The current repository supports synthetic evidence and a small multivariate illustration, not a structural macroeconomic application. The scientific claim is therefore limited to the one the evidence can bear: recurrence diagnostics recover dimensions of stress persistence and organization that are not exhausted by volatility summaries alone.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and identifies the specific gap addressed here. Section 3 introduces the probabilistic framework and the estimands. Section 4 describes the estimators and the implementation logic. Section 5 develops the cluster-adjusted stress functional and states basic theoretical results. Section 6 presents synthetic evidence from the generated artifacts. Section 7 discusses limitations and scope.

2 Literature Review

2.1 Extreme values under dependence

The first strand is the theory of extremes for dependent sequences. Classical presentations of EVT often begin with independent observations because independence exposes the basic tail mechanisms cleanly, but that simplification is rarely adequate for economic time series, where temporal dependence is often strongest precisely in turbulent periods. Leadbetter, Lindgren, and Rootzén [12] provided the foundational framework for extremes under dependence, including the conditions under which clustering enters the limiting behavior of maxima and exceedances. Embrechts, Klüppelberg, and Mikosch [7] and Coles [6] extended that framework into the language of applied risk analysis, making clear that one cannot infer the meaning of rare events from marginal tails alone. For the present paper, the lesson is specific: once dependence is admitted, the number of threshold exceedances no longer exhausts what must be said about stress. Their temporal arrangement becomes part of the estimand.

2.2 Extremal index and clustering diagnostics

The second strand concerns the extremal index and the practical problem of quantifying clustering. O'Brien [14] established the extremal index as a scalar summary of local tail dependence for stationary sequences, thereby giving formal expression to the distinction between isolated and bunched extremes. Later work by Ferro and Segers [9] and Süveges [15] developed estimation strategies that made the object empirically usable, while also underscoring the fragility induced by threshold choice, finite samples, and declustering rules. This literature is indispensable for the present paper, although I do not attempt to improve it on its own terms. My interest lies instead in a more applied question: how should clustering diagnostics be incorporated into stress analysis so that they do not remain isolated technical appendices to otherwise conventional volatility studies? The answer pursued here is to treat runs-based clustering not as a side calculation, but as one ingredient of a broader recurrence-based description of stress.

2.3 Recurrence and return-time analysis

The third strand studies recurrence itself, especially through hitting times and return-time distributions. In dynamical-systems and stochastic-process settings, these quantities have long been understood as structurally meaningful rather than merely descriptive. Hirata, Saussol, and Vaienti [11] showed how return-time statistics encode recurrence properties in a general framework, while Freitas, Freitas, and Todd [10] and Lucarini et al. [13] deepened the connection between rare events, recurrence, and temporal dependence. What matters for the present argument is not the full technical range of that literature, but its conceptual insistence that the spacing of rare events carries information distinct from their marginal severity. Applied econometrics has only partially absorbed that point. Return times are often discussed informally in crisis narratives, yet they are much less often estimated and reported as formal objects inside empirical stress workflows.

2.4 Volatility econometrics and tail-risk baselines

The fourth strand is the econometric baseline against which the present framework should be understood. ARCH and GARCH models [8, 4] established time-varying volatility as a central empirical object, and they remain indispensable whenever the aim is to characterize changing second-moment conditions. Tail-risk measures and conditional downside summaries, including CoVaR [1], extend that baseline toward questions of severity and dependence in bad states. None of this is in dispute. The issue is simply that these tools answer a different question from the one pursued here. They are designed to measure scale, exposure, and tail losses, whereas recurrence diagnostics are designed to measure the temporal organization of repeated exceedances. A period of elevated volatility may contain a single shock, a sequence of loosely related shocks, or a tightly clustered stress episode. Treating those cases as equivalent is often a stronger modeling choice than the econometric discussion acknowledges.

2.5 Macro-financial stress monitoring and systemic-risk measurement

The fifth strand is the literature on stress monitoring and systemic risk. Surveys such as Bisias et al. [3] make clear that this literature is already rich in market-based, balance-sheet-based, and network-oriented indicators. Measures such as CoVaR [1], SRISK [5], and systemic expected shortfall [2] are designed to quantify the severity of distress, the capital shortfall associated with it, and the extent to which risk propagates across institutions. Those are major achievements, and the present paper is not offered as a competitor to them. It should instead be read as targeting a narrower diagnostic dimension that those measures do not make central, namely the recurrence profile of stress through time. The question here is not how much capital is at risk in a bad state, but how stress episodes reappear, bunch together, and persist once threshold-based distress has begun.

2.6 Gap and contribution

Taken together, these literatures leave a surprisingly specific opening. The missing piece is not theory about dependent extremes, nor technique for volatility or systemic-risk measurement. It is an analytic framework in which recurrence, clustering, and return-time structure are treated as explicit stress objects and are estimated within the same reproducible workflow as conventional econometric summaries. That is the contribution pursued in this paper. I formulate a recurrence-based stress framework in probabilistic terms, implement it through an artifact-generating computational pipeline, and use it to motivate a cluster-adjusted stress functional that preserves incidence information while penalizing temporal concentration of extremes. The result is not a new general theory of stress episodes, but a more coherent way to connect established theory on dependent extremes to the practical language of econometric stress analysis.

3 Probabilistic Framework

Let $\{X_t\}_{t\in\mathbb{Z}}$ be a strictly stationary real-valued process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with marginal distribution function F. The object X_t may represent a transformed return, a stress indicator, or another scalar measure whose upper tail corresponds to more severe conditions. The multivariate extension is introduced below.

For a threshold u, define the exceedance indicator

$$I_t(u) = \mathbf{1}\{X_t > u\}.$$

The exceedance probability is

$$p(u) = \mathbb{P}(X_0 > u).$$

In finite samples of length n, high-threshold analysis typically works with a threshold sequence u_n satisfying

$$n\,\overline{F}(u_n) \to \tau \in (0, \infty),$$

where $\overline{F}(u) = 1 - F(u)$. This keeps exceedances rare but observable.

If $t_1 < \cdots < t_{N_n}$ denote the exceedance times among $\{1, \ldots, n\}$, the return times are

$$\tau_{j,n} = t_{j+1} - t_j, \qquad j = 1, \ldots, N_n - 1.$$

These random intervals summarize recurrence to the stress region. Their empirical distribution captures spacing, dispersion, and asymmetry in a way that event counts cannot.

To describe clustering, fix a run parameter $r \geq 1$. A new cluster begins at time t if $X_t > u_n$ and the previous r observations do not exceed u_n. The corresponding finite-level cluster-start indicator is

$$C_{t,n}^{(r)} = I_t(u_n) \prod_{j=1}^{r} \left(1 - I_{t-j}(u_n)\right).$$

The associated finite-level clustering coefficient is

$$\theta_n^{(r)} = \frac{\mathbb{E}[C_{0,n}^{(r)}]}{\mathbb{E}[I_0(u_n)]} = \mathbb{P}(X_{-1} \leq u_n, \ldots, X_{-r} \leq u_n \mid X_0 > u_n).$$

When the standard runs approximation is appropriate, $\theta_n^{(r)}$ is a finite-level analogue of the extremal index: lower values indicate stronger local clustering of exceedances.

For a d-variate process $\{X_{t,j}\}$ with component-specific thresholds $u_{j,n}$, define

$$I_{t,j}(u_{j,n}) = \mathbf{1}\{X_{t,j} > u_{j,n}\}, \qquad S_t = \frac{1}{d}\sum_{j=1}^{d} I_{t,j}(u_{j,n}).$$

The scalar $S_t \in [0, 1]$ is a multivariate stress-state index: it records the share of active stress coordinates at time t. It is descriptive, not causal.

4 Estimators and Implementation

4.1 Sample estimators

Given observations $X_1, \dots, X_n$, the empirical exceedance rate at threshold u_n is

$$\widehat{p}_n(u_n) = \frac{1}{n} \sum_{t=1}^{n} I_t(u_n).$$

The runs-based estimator of the finite-level clustering coefficient is

$$\widehat{\theta}_n^{(r)} = \frac{\sum_{t=r+1}^{n} C_{t,n}^{(r)}}{\sum_{t=1}^{n} I_t(u_n)},$$

whenever the denominator is positive. This estimator is simple, transparent, and computationally appropriate for a reproducible pipeline. It is not always the most statistically efficient choice, but it is easy to audit and aligns with the article's emphasis on inspectable artifacts.

The empirical return-time distribution is

$$\widehat{G}_n(x) = \frac{1}{N_n - 1} \sum_{j=1}^{N_n - 1} \mathbf{1}\{\tau_{j,n} \le x\},$$

provided $N_n \ge 2$. Summary functionals such as the mean, median, upper quantiles, and maximum of $\{\tau_{j,n}\}$ are used throughout the evidence section.

For the multivariate setting, the empirical stress-state index is

$$\widehat{S}_t = \frac{1}{d} \sum_{j=1}^{d} I_{t,j}(u_{j,n}),$$

and joint stress is defined by the event $\{\widehat{S}_t \ge 2/d\}$ in the repository's current illustration. This threshold can be varied; the key point is to make the rule explicit.

4.2 Threshold and run-length choices

Threshold selection is a substantive modeling decision, not a clerical preprocessing step. A threshold set too low destroys the rare-event interpretation; a threshold set too high produces severe finite-sample instability. The repository therefore treats threshold sensitivity as part of the diagnostic output rather than as a hidden implementation detail. In applied work, one should report how clustering summaries move across a threshold grid and how the chosen run length r affects cluster identification.

The run parameter r has an equally direct interpretation. Larger values merge nearby exceedances into common clusters and therefore tend to increase measured persistence. This does not make large r "better." It means that r encodes what the analyst wishes to count as persistence. A serious stress paper should disclose that choice and test its sensitivity.

4.3 Repository implementation

The repository generates figure and CSV artifacts through a deterministic script that proceeds from raw or simulated series to threshold definition, recurrence summaries, baseline econometric diagnostics, and article-facing outputs. The conceptual pipeline contains eight recorded stages: raw series, transformations, stress-region definition, exceedance events, clustering diagnostics, return-time diagnostics, econometric baselines, and article evidence. This separation matters because recurrence statistics are sensitive to thresholding and clustering rules; saved intermediate tables are therefore part of the method.

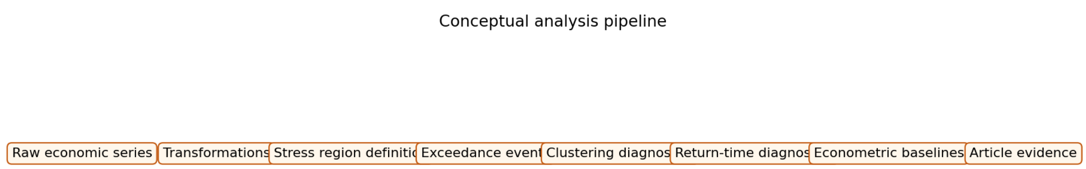

Figure 1: Conceptual pipeline for the repository-native workflow. The methodological point is that recurrence objects are treated as explicit data products rather than informal visual impressions.

5 A Cluster-Adjusted Stress Functional

5.1 Definition

The paper's original theoretical contribution is deliberately narrow, but it is intended to be more than notational decoration. If recurrence is to be treated as part of stress measurement rather than as an auxiliary descriptive remark, then clustering information must be translated into a quantity that can be compared across processes and thresholding schemes. The functional introduced here is designed for exactly that purpose:

$$\Lambda(u_n, r) = \frac{p(u_n)}{\theta_n^{(r)}}.$$

Its sample analogue is

$$\widehat{\Lambda}_n(u_n, r) = \frac{\widehat{p}_n(u_n)}{\widehat{\theta}_n^{(r)}},$$

whenever $\widehat{\theta}_n^{(r)} > 0$.

The interpretation is straightforward but worth stating carefully. The exceedance rate $\widehat{p}_n$ records how often the process enters the stress region, while the clustering coefficient $\widehat{\theta}_n^{(r)}$ records how isolated those exceedances are once they occur. Dividing the first quantity by the second therefore leaves incidence unchanged in the benchmark case of no clustering and increases the measured stress load when exceedances arrive in tighter temporal groups. Because $\theta_n^{(r)} \leq 1$, the functional always satisfies $\Lambda(u_n, r) \geq p(u_n)$. In that sense, it may be read as a cluster-adjusted stress load: identical event frequencies need not imply identical stress severity if one process concentrates those events into more persistent bursts.

The simplicity of the definition is intentional. The aim is not to compete with the richer theory of tail processes or extremal dependence structures, but to produce a scalar summary that remains mathematically interpretable while being usable inside an applied stress workflow.

5.2 Basic properties

Proposition 1 (Monotone ordering by clustering)**.** *Fix a threshold u and run length r. Consider two stationary processes A and B such that*

$$p_A(u) = p_B(u) = p(u) > 0, \qquad \theta_A^{(r)}(u) < \theta_B^{(r)}(u) \leq 1.$$

Then

$$\Lambda_A(u, r) > \Lambda_B(u, r).$$

Proof sketch. The conclusion follows immediately from the definition $\Lambda(u, r) = p(u)/\theta^{(r)}(u)$ and positivity of $p(u)$. For equal exceedance probability, the process with smaller clustering coefficient has the larger cluster-adjusted stress load. The content of the proposition is interpretive rather than algebraically deep: equal event frequency should not imply equal stress severity when one process exhibits stronger local bunching of extremes.

Proposition 2 (Independent benchmark)**.** *If exceedances are temporally independent at threshold u, then for every $r \geq 1$,*

$$\theta^{(r)}(u) = 1 \qquad \textit{and} \qquad \Lambda(u, r) = p(u).$$

Proof sketch. Under independence,

$$\mathbb{P}(X_{-1} \leq u, \ldots, X_{-r} \leq u \mid X_0 > u) = 1,$$

so the finite-level clustering coefficient equals one. The cluster adjustment then disappears. This proposition supplies the correct benchmark: the functional adds information only when temporal dependence in extremes is present.

Proposition 3 (Plug-in consistency)**.** *Assume:*

(A1) *$\{X_t\}$ is strictly stationary and strongly mixing with coefficients sufficient for a law of large numbers for the exceedance and cluster-start indicators;*

(A2) *u_n is a high-threshold sequence such that $p(u_n) > 0$ for all n and $p(u_n) \to 0$;*

(A3) *for a fixed run length r, $\theta_n^{(r)} \to \theta^{(r)} \in (0, 1]$;*

(A4) *$\widehat{p}_n(u_n) \xrightarrow{p} p(u_n)$ and $\widehat{\theta}_n^{(r)} \xrightarrow{p} \theta^{(r)}$.*

Then

$$\widehat{\Lambda}_n(u_n, r) - \frac{p(u_n)}{\theta^{(r)}} \xrightarrow{p} 0.$$

If, in addition, $p(u_n) \to p^\star$ *and* $\theta^{(r)} \to \theta^\star > 0$*, then*

$$\widehat{\Lambda}_n(u_n, r) \xrightarrow{p} \frac{p^\star}{\theta^\star}.$$

Proof sketch. Under the stated assumptions, the numerator and denominator converge in probability to positive limits. The continuous mapping theorem applied to $(x, y) \mapsto x/y$ on $\mathbb{R} \times (0, \infty)$ gives the result. The proposition is intentionally conservative: it does not claim new asymptotic theory for $\widehat{\theta}_n^{(r)}$, only that once consistent ingredients are available, the plug-in stress functional is itself consistent.

Proposition 4 (Run-length monotonicity)**.** *Fix a threshold* u_n *and two run lengths* $1 \le r_1 < r_2$*. Then*

$$\theta_n^{(r_2)} \le \theta_n^{(r_1)}.$$

Consequently, whenever both quantities are positive,

$$\Lambda(u_n, r_2) \ge \Lambda(u_n, r_1).$$

Proof sketch. By definition,

$$\theta_n^{(r)} = \mathbb{P}(X_{-1} \le u_n, \dots, X_{-r} \le u_n \mid X_0 > u_n)\,.$$

If $r_2 > r_1$, then the event

$$\{X_{-1} \le u_n, \dots, X_{-r_2} \le u_n\}$$

is a subset of

$$\{X_{-1} \le u_n, \dots, X_{-r_1} \le u_n\}.$$

Conditional probabilities therefore satisfy $\theta_n^{(r_2)} \le \theta_n^{(r_1)}$. Since $\Lambda(u_n, r) = p(u_n)/\theta_n^{(r)}$ with fixed numerator, the second claim follows. The interpretation is immediate: increasing the run length makes the clustering rule stricter, which can only weakly increase the cluster-adjusted stress load.

Proposition 5 (Finite-sample perturbation bound)**.** *Let* $p > 0$ *and* $\theta > 0$*, and let* $\widetilde{p}$ *and* $\widetilde{\theta}$ *be estimators such that*

$$|\widetilde{p} - p| \le \varepsilon_p, \qquad |\widetilde{\theta} - \theta| \le \varepsilon_\theta < \theta.$$

Then

$$\left|\frac{\widetilde{p}}{\widetilde{\theta}} - \frac{p}{\theta}\right| \le \frac{\varepsilon_p}{\theta - \varepsilon_\theta} \frac{p\,\varepsilon_\theta}{\theta(\theta - \varepsilon_\theta)}.$$

Proof sketch. Write

$$\frac{\widetilde{p}}{\widetilde{\theta}} - \frac{p}{\theta} = \frac{\theta(\widetilde{p} - p) + p(\theta - \widetilde{\theta})}{\theta\widetilde{\theta}}.$$

Taking absolute values and using $\widetilde{\theta} \geq \theta - \varepsilon_\theta > 0$ gives

$$\left|\frac{\widetilde{p}}{\widetilde{\theta}} - \frac{p}{\theta}\right| \leq \frac{\theta\varepsilon_p + p\varepsilon_\theta}{\theta(\theta - \varepsilon_\theta)},$$

which is the stated bound after separating terms. The interpretation is important: estimation error in the clustering term is amplified when θ is small, so cluster-adjusted stress is most numerically delicate precisely in the regimes where clustering is strongest.

Remark 1. *The functional $\Lambda(u_n, r)$ is best understood as a ranking and summary device, not as a welfare object or a structural sufficient statistic. Its value lies in comparative stress characterization.*

Remark 2. *The perturbation bound explains why threshold sensitivity cannot be treated as a cosmetic appendix item. If threshold changes move $\widehat{\theta}_n^{(r)}$ toward zero, the same sampling error in the clustering estimator induces a larger error in $\widehat{\Lambda}_n$. This is a concrete mathematical reason to report threshold and run-length robustness.*

6 Synthetic Evidence

6.1 Core numerical outputs

Table 1 collects the generated quantities used in the analysis.

Table 1: Core artifact summaries

Object	Numerical summary
Extremal index by series	Clustered synthetic stress: 0.673; isolated synthetic stress: 0.780; logistic benchmark observable: 1.000
Return-time distribution	249 return times; mean 20.1 periods; median 14.0; 90th percentile 40.2; maximum 90
Multivariate stress timeline	120 dated observations; 12 joint exceedance dates; maximum of 2 active series; mean stress score 0.150
Volatility comparison	Mean rolling volatility during exceedances 0.3429; outside exceedances 0.2671; exceedance count 90

These quantities do not appear here as decorative summaries appended to a computational pipeline. They define the evidentiary boundary of the paper. Every substantive claim that follows is tied either to these numerical outputs or to the saved source tables from which the corresponding figures are drawn. That restriction is methodologically important, because the manuscript is intended to argue from inspectable evidence rather than from narrative inflation.

6.2 Synthetic clustering benchmarks

The clearest evidence for the paper's thesis comes from the extremal-index comparison between the two synthetic stress benchmarks. The clustered benchmark yields an estimated extremal index of

0.673, whereas the more isolated benchmark yields 0.780. The numerical gap is not large enough to justify dramatic rhetoric, and the manuscript should resist the temptation to manufacture significance through tone. Its importance lies elsewhere. The ranking shows that recurrence diagnostics recover a difference in temporal organization that would be awkward to express, and even easier to ignore, if the discussion remained confined to generic language about instability or turbulence.

This is also the context in which the cluster-adjusted functional acquires substantive meaning rather than remaining a formal convenience. For comparable exceedance incidence, the series with the smaller clustering coefficient receives the larger stress load, because the same overall amount of tail activity is being delivered in a more temporally concentrated way. The theoretical section is therefore not separate from the synthetic evidence; it explains why the empirical ranking should matter in the first place.

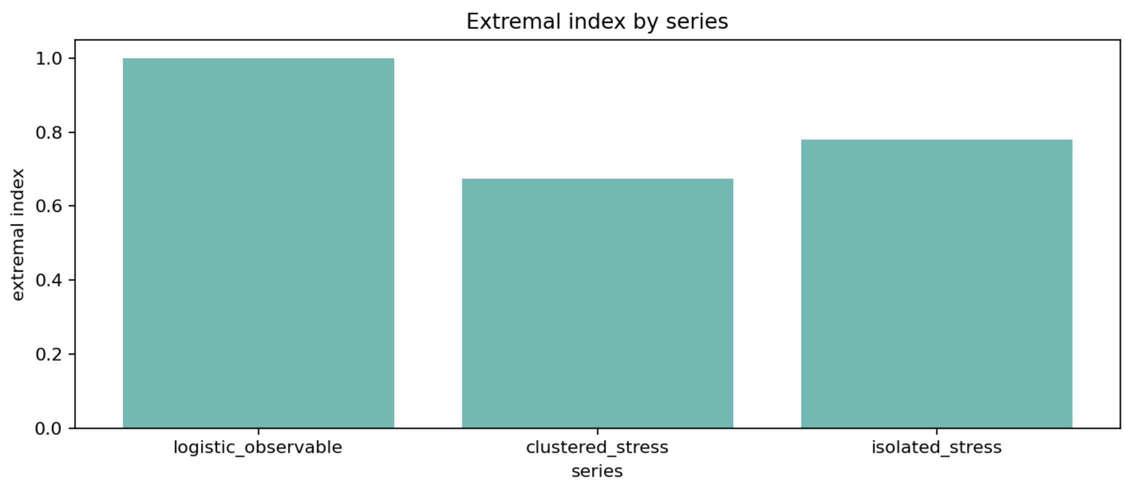

Figure 2: Estimated extremal indices across benchmark series. Lower values indicate stronger local clustering of exceedances. The result is finite-sample and threshold-dependent, but it recovers the intended ranking between clustered and more isolated stress processes.

6.3 Threshold robustness and uncertainty

Any recurrence-based stress diagnostic inherits two forms of uncertainty at once. The first is ordinary sampling uncertainty: finite samples generate noisy exceedance counts, noisy cluster counts, and therefore noisy clustering summaries. The second is design uncertainty: the reported statistic depends on threshold choice and, for runs-based procedures, on the analyst's declustering rule. For that reason, robustness over a threshold grid is not a cosmetic appendix exercise but part of the object being measured.

The repository's threshold-sensitivity artifact makes this point directly. For the benchmark observable used in the threshold scan, the runs-based clustering estimate is 0.888 at the 0.90 quantile with 500 exceedances and rises to 0.909 at the 0.93 quantile with 350 exceedances. By the 0.95 quantile, the estimate reaches 1.000 with 250 exceedances and remains at that level through the 0.97 and 0.99 quantiles, even as the number of exceedances falls to 150 and then 50. The diagnostic message is clear: raising the threshold reduces the apparent evidence of clustering in this benchmark, but it does so while rapidly thinning the sample on which the estimate rests. Stability at very high thresholds is therefore not self-validating; it may reflect both genuine tail behavior and the loss of finite-sample resolution.

This threshold path also clarifies what uncertainty quantification should mean in a completed empirical application. A single point estimate of $\widehat{\theta}_n^{(r)}$ or $\widehat{\Lambda}_n(u_n, r)$ is not enough, because inferential uncertainty and threshold sensitivity interact. A serious applied version of the framework should report block-bootstrap or subsampling-based uncertainty measures over a threshold grid, so that the analyst can see not only whether the estimate is noisy at a fixed cutoff, but also whether the substantive ranking survives reasonable perturbations of the stress definition itself.

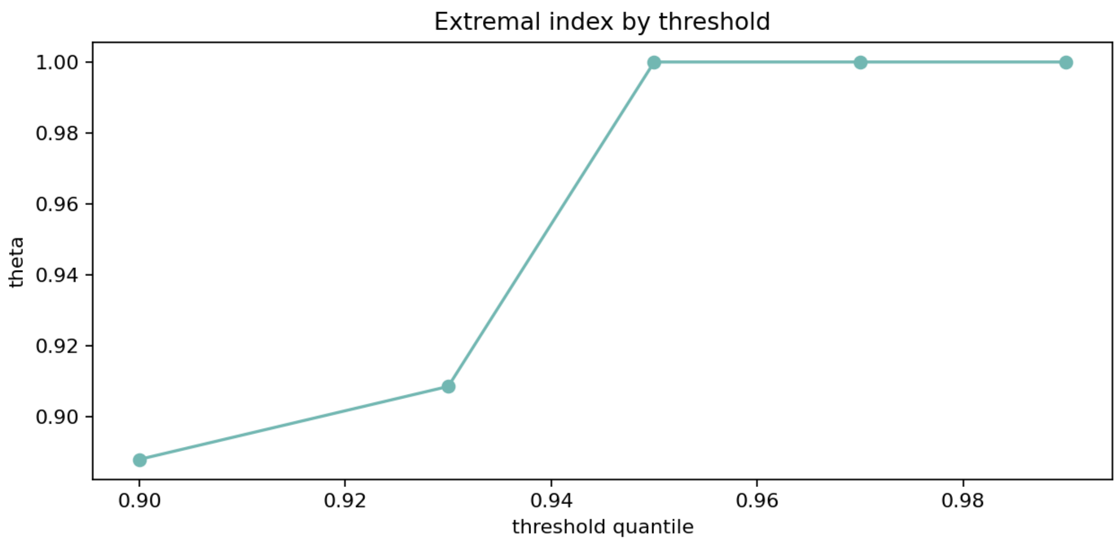

Figure 3: Threshold sensitivity of the clustering estimate. The figure is useful not because it identifies one “correct” cutoff, but because it shows how clustering diagnostics move as the threshold becomes more selective and the effective sample of exceedances shrinks.

6.4 Return-time dispersion

The return-time artifact contains 249 inter-exceedance intervals with mean 20.1, median 14.0, 90th percentile 40.2, and maximum 90. What matters here is not simply that the intervals vary, since variation is inevitable in any finite sample, but that the variation is large enough to make a single summary interval misleading. Two stress processes can produce roughly comparable frequencies of exceedance while differing substantially in the distribution of quiet periods that separate those exceedances. Once that possibility is admitted, event counts and volatility spikes cease to be sufficient descriptions of persistence.

The gap between the mean and the median reinforces this point. Recurrence is not well captured by a single “typical” waiting time, and any empirical discussion that reduces stress persistence to one average duration risks mistaking a distributional phenomenon for a scalar one. In this sense, return-time analysis does not merely complement threshold exceedance counts; it alters what a serious description of stress is required to include.

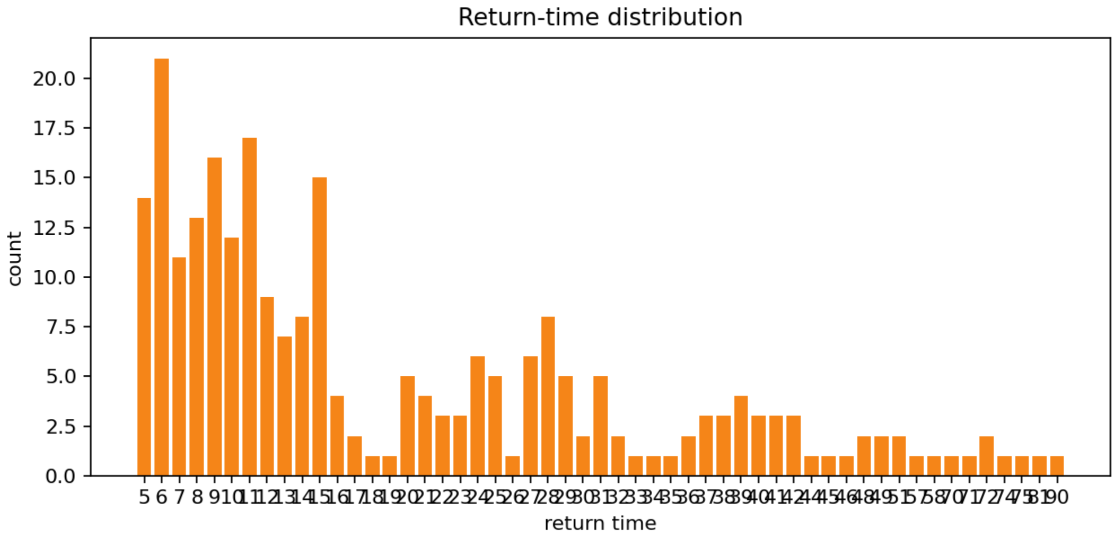

Figure 4: Empirical distribution of return times between exceedances. The wide spread confirms that recurrence is a distributional feature, not a single-duration label.

6.5 Multivariate stress activation

The multivariate illustration records 120 observations and 12 joint exceedance dates, with at most two simultaneously active series and a mean stress score of 0.150. The claim supported by these numbers is intentionally modest, but it is still consequential. Stress is not uniformly distributed across coordinates through time: some episodes are effectively univariate, while others involve simultaneous activation across variables. That descriptive distinction should be measured before stronger language about contagion, propagation, or systemic amplification is introduced.

The stress-state index is therefore best understood as a screening device for joint activation rather than as a substitute for structural multivariate analysis. Its purpose is to mark the temporal geometry of shared stress, not to explain why that geometry arises.

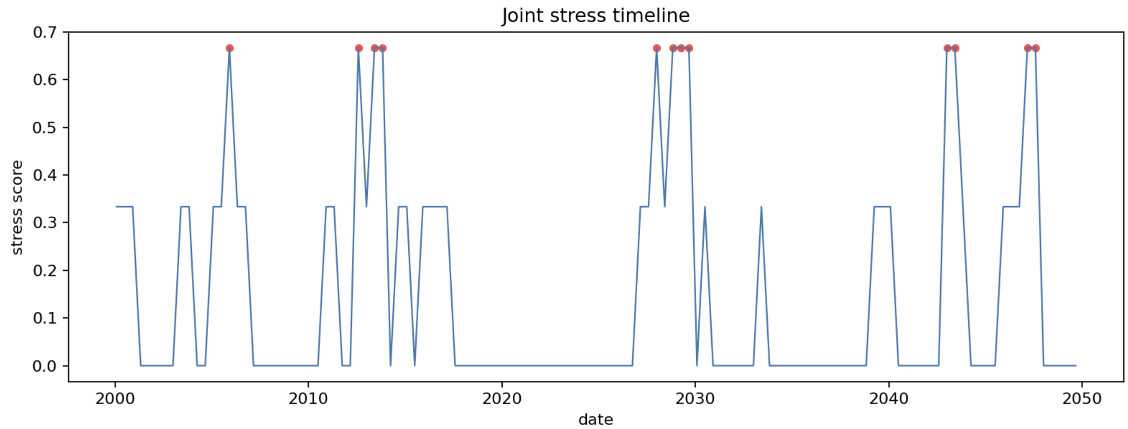

Figure 5: Multivariate stress-state timeline. The figure does not identify transmission channels; it records when stress is isolated and when it is joint across observables.

6.6 Why the volatility baseline remains necessary

The repository's baseline comparison shows higher mean rolling volatility during exceedance periods (0.3429) than outside them (0.2671). This is both substantively useful and entirely unsurprising. A recurrence framework should not be judged by whether it overturns established volatility evidence, because that is not the standard it is meant to meet. Its purpose is to recover information that volatility summaries do not encode directly.

The proper conclusion is therefore one of complementarity rather than rivalry. Volatility remains the natural object when the question concerns changing scale conditions, while recurrence diagnostics become essential when the question concerns how extreme states reappear and bunch together through time. A stress workflow that excludes either object is, in different ways, incomplete.

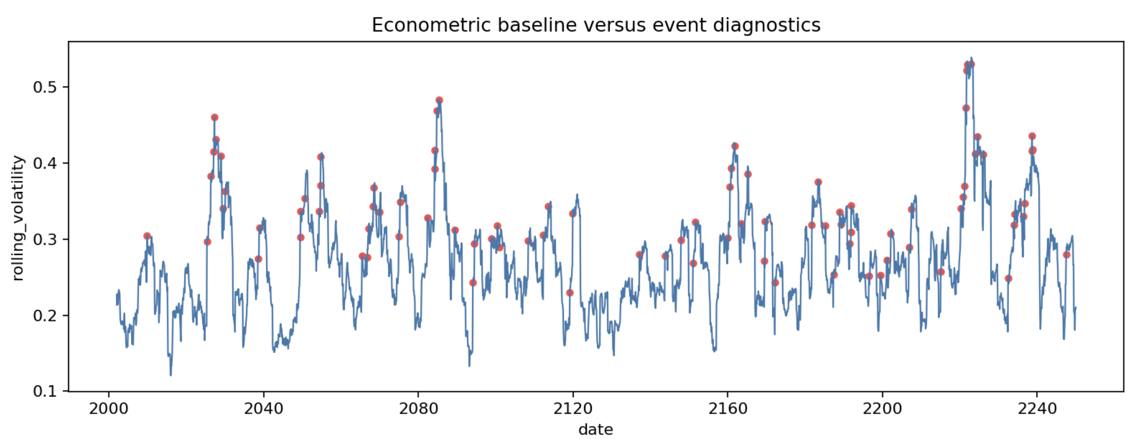

Figure 6: Baseline econometric and recurrence-oriented diagnostics are complementary. Elevated volatility during exceedance periods is informative, but it does not encode clustering intensity or return-time dispersion.

6.7 A minimal real-data illustration

The repository currently contains only a small didactic raw-data example rather than a full empirical panel, so the following illustration should be read as proof of concept rather than as substantive macro-financial evidence. Even so, it helps show how the recurrence language behaves once one leaves the purely synthetic setting. Using the example inflation series bundled with the repository, an upper-tail threshold at the 0.80 quantile produces five exceedances, all concentrated in a single run from August 2021 through December 2021. The corresponding empirical picture is not one of scattered inflation surprises, but of one compact inflation-stress episode.

The equity example becomes informative only after an economically meaningful transformation. On raw levels, an upper-tail threshold would identify booming prices rather than stress. If one instead studies upper-tail equity losses obtained by sign-flipping monthly log returns, the same 0.80 quantile rule yields five exceedances distributed across five separate clusters, spanning March 2020 to September 2021. In this toy example, inflation stress is therefore persistent once active, whereas equity-loss stress is more episodic. The two series also overlap in only one joint stress month, September 2021, under this coarse thresholding rule.

This illustration is intentionally limited. The sample is short, the thresholds are chosen for didactic transparency rather than empirical optimality, and no inference should be attached to the resulting counts. Its value is narrower and more methodological: it shows that the recurrence framework can articulate distinctions between persistence and intermittency even in a minimal real-data setting, and that those distinctions are already more informative than a generic statement that both series were "under stress" at some point.

7 Limitations and Scope

The limits of the paper are easiest to understand once its contribution is stated without inflation. The theoretical results are intentionally narrow. They establish interpretable properties of the cluster-adjusted stress functional and show how its behavior depends on the clustering term, but they do not amount to a new asymptotic theory for extremal-index estimation, a solution to threshold selection, or a general finite-sample theory for recurrence diagnostics. The paper therefore contributes an organizing functional and a set of elementary results around it, not a replacement for the deeper probabilistic literature on dependent extremes.

The empirical limits are equally important. The evidence presented here is primarily synthetic, which is appropriate for showing what the diagnostics do and do not measure, but insufficient for strong claims about macro-financial systems in the wild. A more mature empirical paper would need defended threshold rules, uncertainty quantification, robustness analysis across run lengths and sample windows, and an interpretation anchored in a specific domain rather than in generic language about stress. Without those additions, the present manuscript should be read as methodological preparation for empirical work rather than as empirical resolution.

The multivariate component is narrower still. The stress-state index records when threshold-based stress is isolated and when it is joint, but it does not identify contagion, directional spillovers, or structural transmission channels. That limitation is not incidental. It marks the boundary between descriptive recurrence diagnostics and the larger enterprise of systemic-risk modeling.

These constraints should not be treated as apologetic caveats appended after the fact. They are part of the paper's intellectual discipline. The contribution is a methods and theory bridge for recurrence diagnostics in stress analysis, and it is more useful to state that contribution clearly than to overstate what the present evidence can bear.

8 Conclusion

The central argument of the paper is that stress analysis must distinguish tail magnitude from tail organization. Econometric and risk-measure literatures have developed powerful tools for the first object, but the second requires a more explicit treatment of recurrence, clustering, and return-time behavior than stress analysis usually receives. The framework developed here is meant to supply that treatment in a form that is mathematically legible and computationally reproducible.

Its contribution is therefore best described in bounded terms. The paper does not propose a new general theory of dependent extremes, nor does it resolve the empirical problems that arise when recurrence diagnostics are taken to real macro-financial data. What it does show is that recurrence can be formalized as a stress object, estimated through transparent pipeline components, summarized by an interpretable cluster-adjusted functional, and read alongside baseline volatility diagnostics without collapsing into them. The synthetic evidence supports that narrower claim, and the value of the paper lies in making the claim precisely enough that later empirical or theoretical work can either build on it or reject it on clear grounds.

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